AINE 312 Term Project
US Traffic Accidents Analysis
Colorado (CO) and Oklahoma (OK)

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The dataset was originally very large, I filtered it to only keep Colorado and Oklahoma since those are my assigned states.

After that, it dropped to around 174k rows. Colorado is a bit higher than Oklahoma, but not by a much.

Each row has things like location, severity, start/end times, weather info, and also some flags like if there is a junction or signal. There is actually a lot of detail in each record.

Missing Value Treatment

There were missing values, mostly in weather columns. I checked a few options here.

Instead of deleting rows, I filled them using the median. It felt safer because there are some extreme values in the dataset.

I didn’t separate it by state. Probably should have, but I kept it simple to avoid overcomplicating things.

In this particular study, I didn't break down the median by state, but it's definitely something to think about for future research, especially since Colorado and Oklahoma have pretty different climates.

Duration

I created a duration column from start and end time.

Some values were negative, which doesn’t really make sense, so I removed those.

Also noticed some very large durations. Like way too large. I treated those as outliers and removed them using IQR. They didn’t look realistic anyway.

Weather conditions had slightly different naming for similar things. I cleaned that a bit, nothing too complex.

Also added: duration (minutes), hour of day

These helped later when I was trying to see patterns.

Severity

Most accidents in Oklahoma are severity 2. That part is very clear.

Colorado is different though. It has more severity 3 and 4 compared to Oklahoma. Not sure exactly why at first, but it stands out.

So already there is a difference between the states.

General Stats

Duration is not consistent. Some are short, some are really long. That makes the average a bit misleading.

Temperature is lower in Colorado most of the time. Could be related to road conditions, especially in winter.

Humidity and visibility looked similar between states. I expected more difference honestly.

Correlation

I checked correlations just to see if anything obvious shows up.

Weather variables relate to each other: Precipitation, visibility (negative). humidity, visibility

But severity didn't really connect strongly to anything alone. So it's probably not something simple.

Weather vs Severity

At first, clear weather has the most accidents. That seems wrong, but actually it's just because more people are driving.

When looking at severity instead of count, bad weather (rain, snow, fog) looks worse. That part makes sense.

Overcast was a bit unexpected. It shows more severity than I thought. Not really sure why. Maybe drivers don't react the same way as in heavy rain.

Road

Junctions seem more dangerous. That part is kind of expected because there is more interaction between cars.

Traffic signals show up a lot too, but I don't think they cause accidents. It's more that intersections are busy areas.

So, it's more about location than the signal itself.

Statistical analysis:

	Severity	Distance(mi)	Temperature(F)	...	Wind_Speed(mph)	Precipitation(in)	Duration_Minutes
count	148050.000000	148050.000000	146478.000000	...	140090.000000	94589.000000	148050.000000
mean	2.256035	0.372192	56.024867	...	9.824465	0.005563	55.184048
std	0.530164	1.994022	22.184894	...	6.237272	0.091262	32.940158
min	1.000000	0.000000	-89.000000	...	0.000000	0.000000	1.216667
25%	2.000000	0.000000	39.000000	...	5.800000	0.000000	29.733333
50%	2.000000	0.000000	57.000000	...	9.000000	0.000000	44.633333
75%	2.000000	0.000000	74.000000	...	13.000000	0.000000	74.733333
max	4.000000	254.399994	111.900000	...	254.300000	24.000000	166.266667

[8 rows x 9 columns]

At the end, Colorado and Oklahoma are not the same in terms of accident severity.

Colorado has more serious cases. Oklahoma is more concentrated in lower severity.

Night accidents are fewer, but they seem worse. Weather also matters, especially rain and snow.

Road features like junctions increase risk. Signals seem to reduce extreme cases a bit.

Overall, nothing is controlled by just one thing. It's more like a combination of factors.

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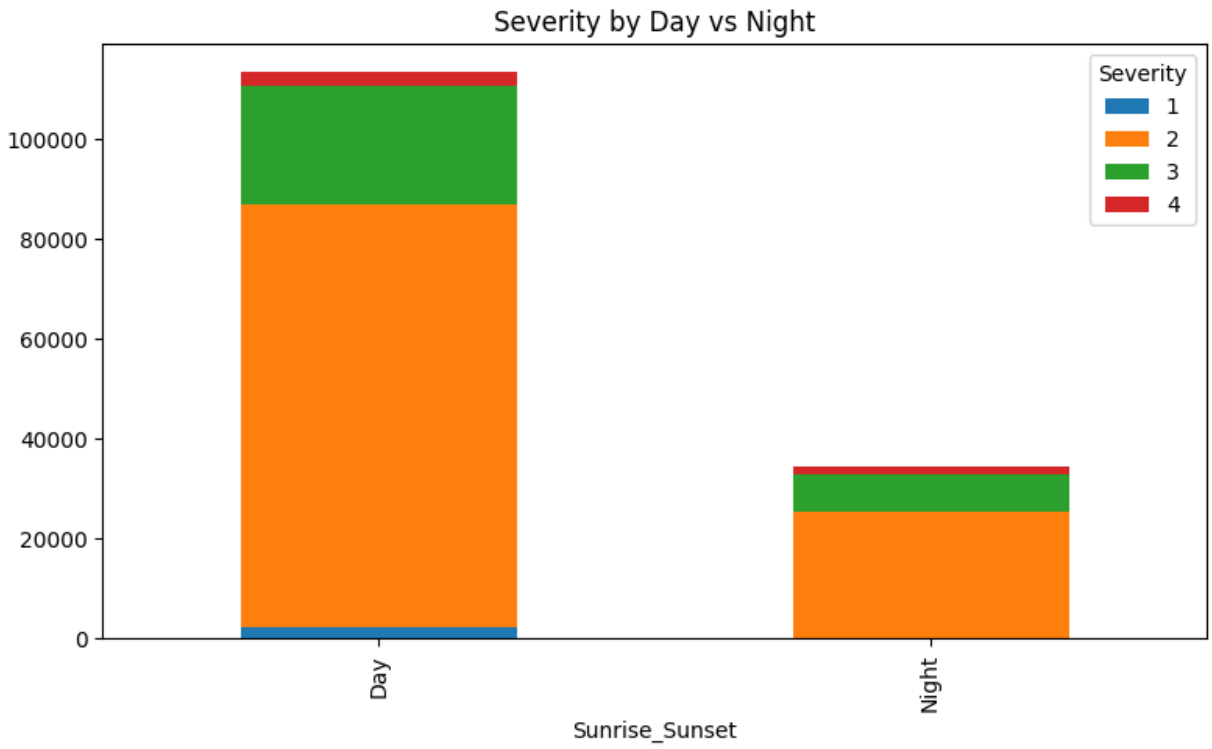
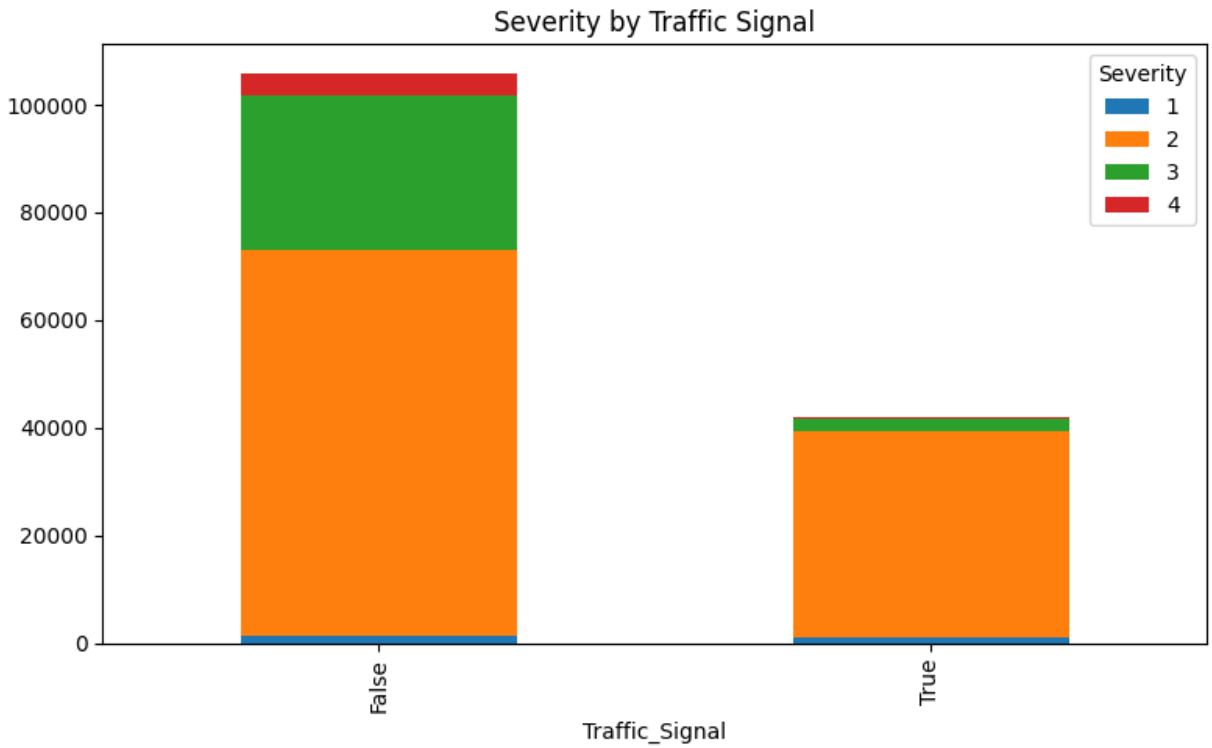
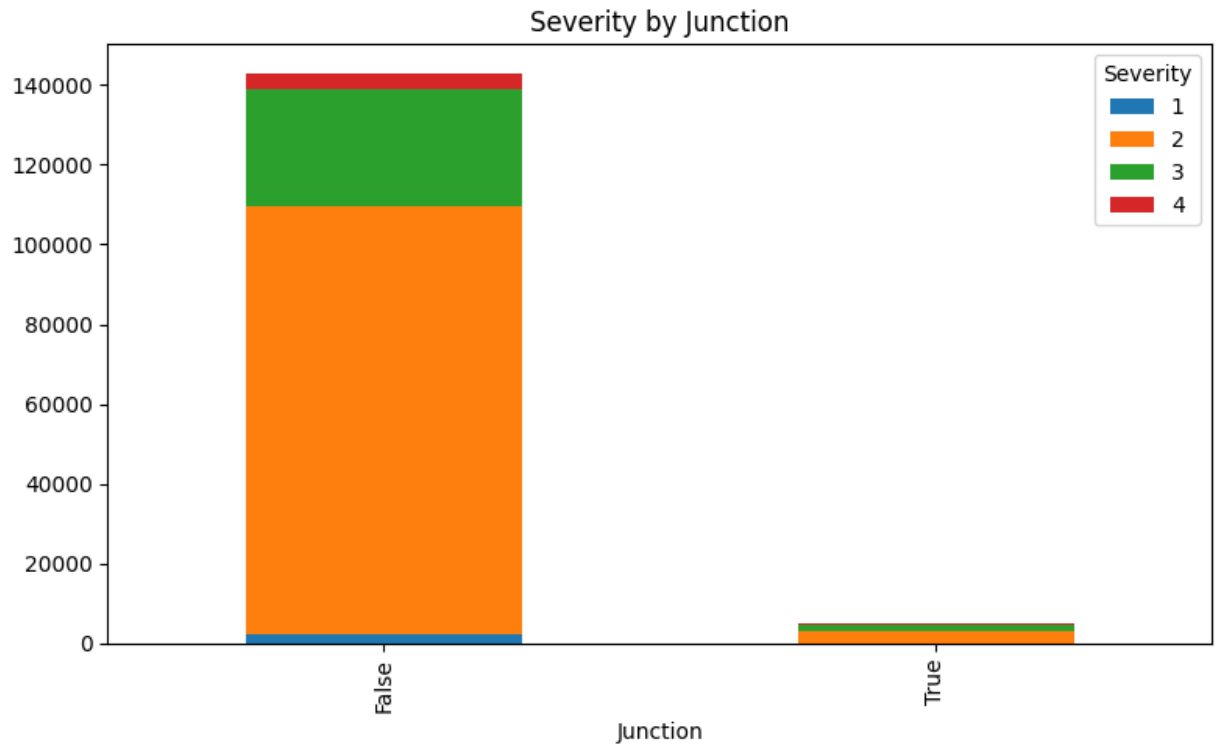


Accidents are three times more likely to occur at day light than at night However, the *proportion* of severity remains remarkably consistent regardless of lighting conditions.



Most of the accidents happen because there are no traffic lights there



The overwhelming majority of accidents occur on standard stretches of road rather than at junctions (Junction = False).

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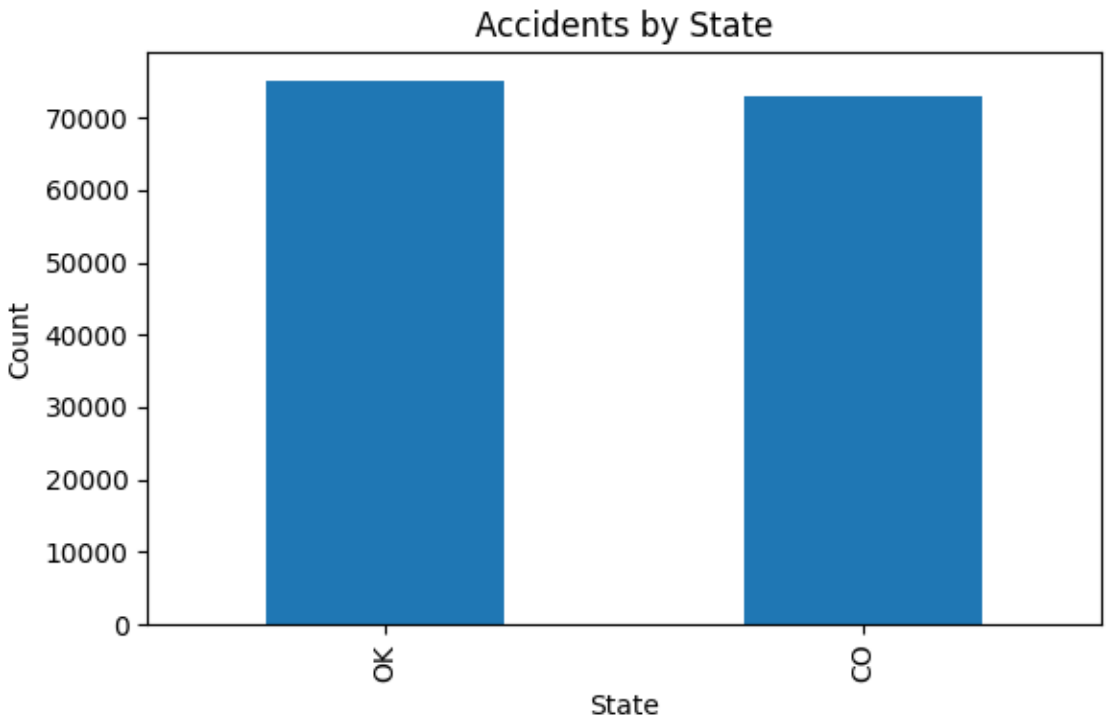
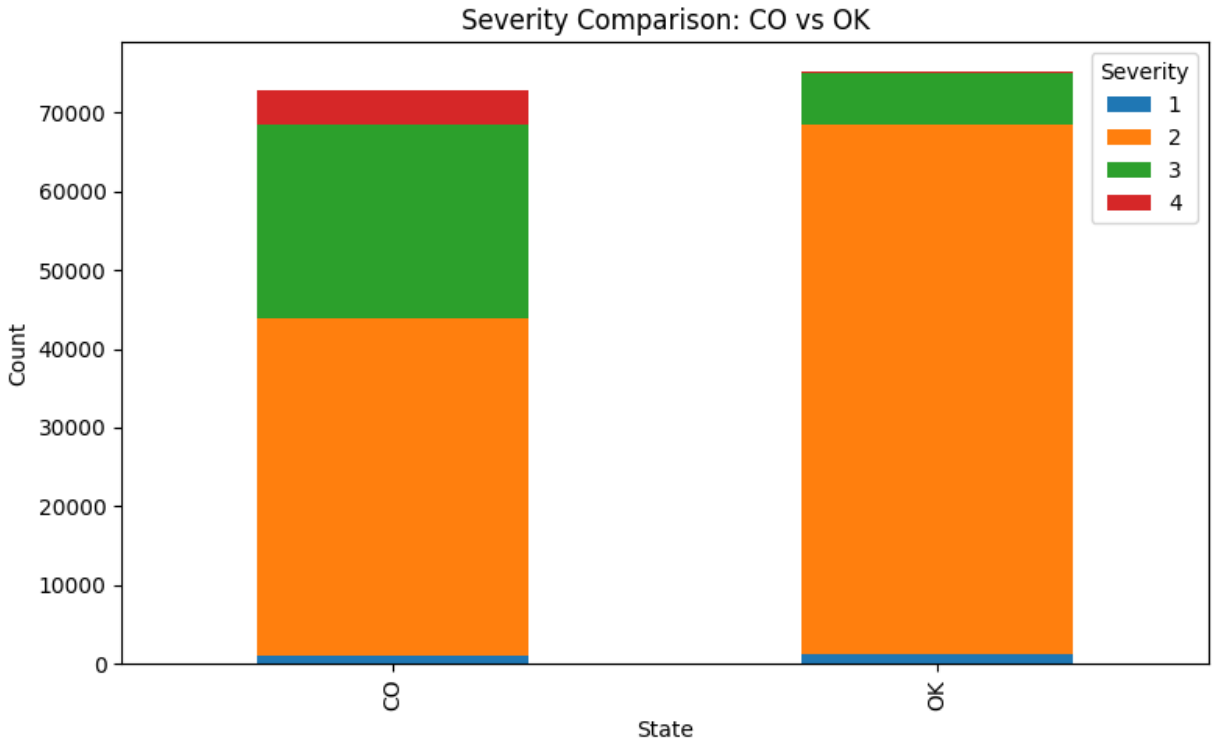
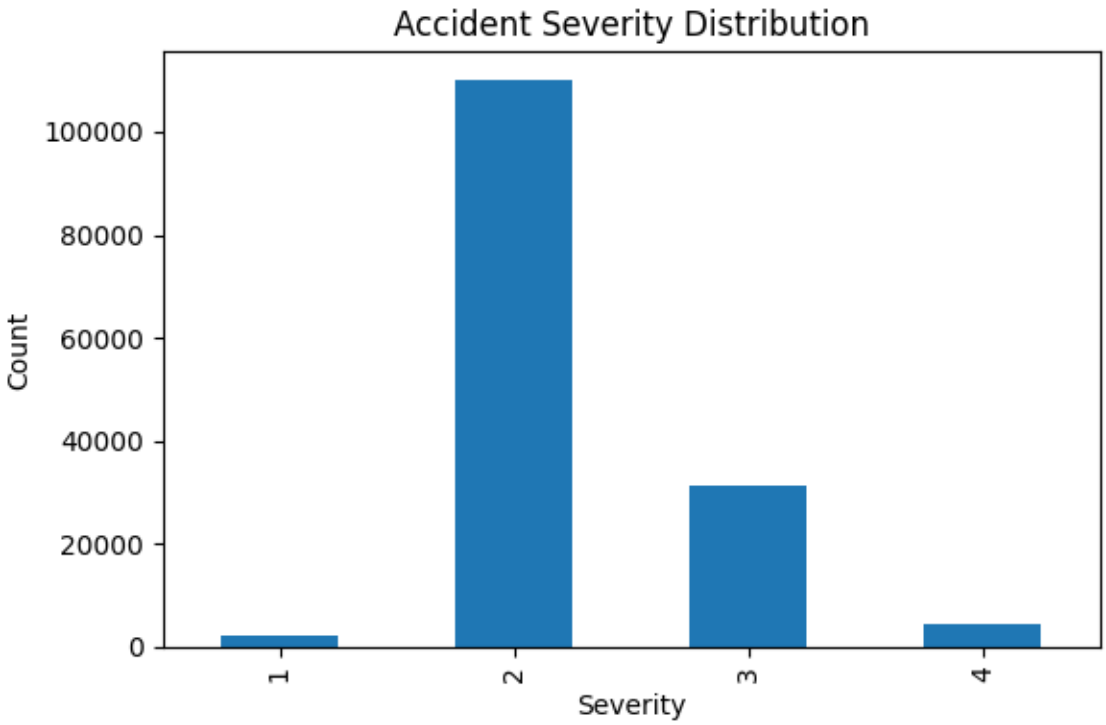


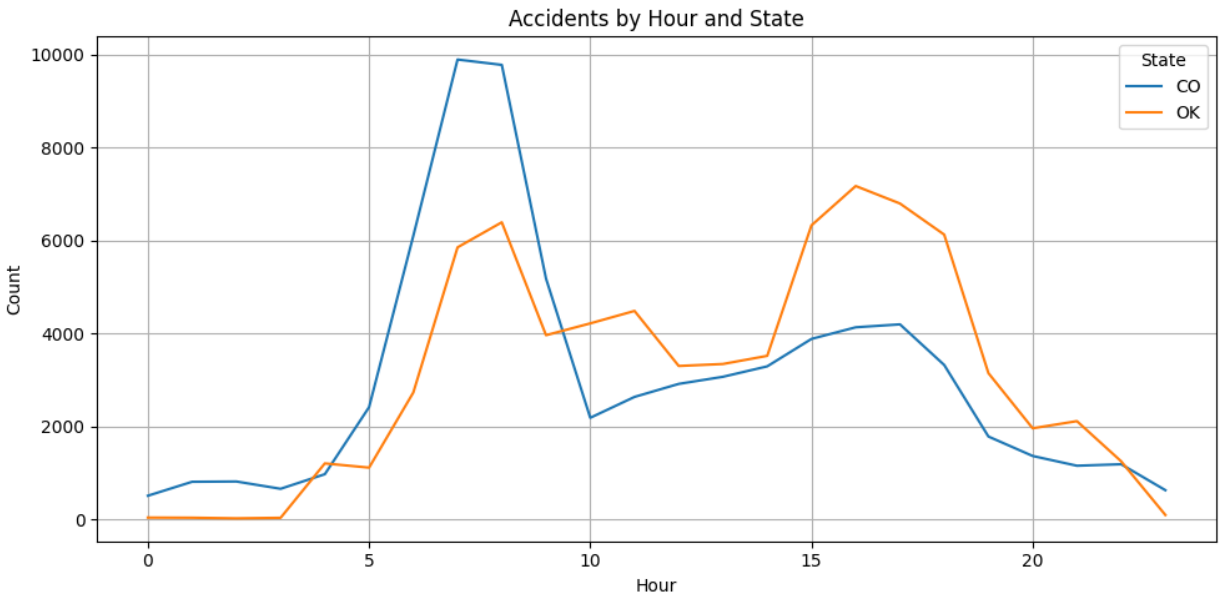
The upper chart shows the Accidents rate in states (CO-OK) each (70000-75000)



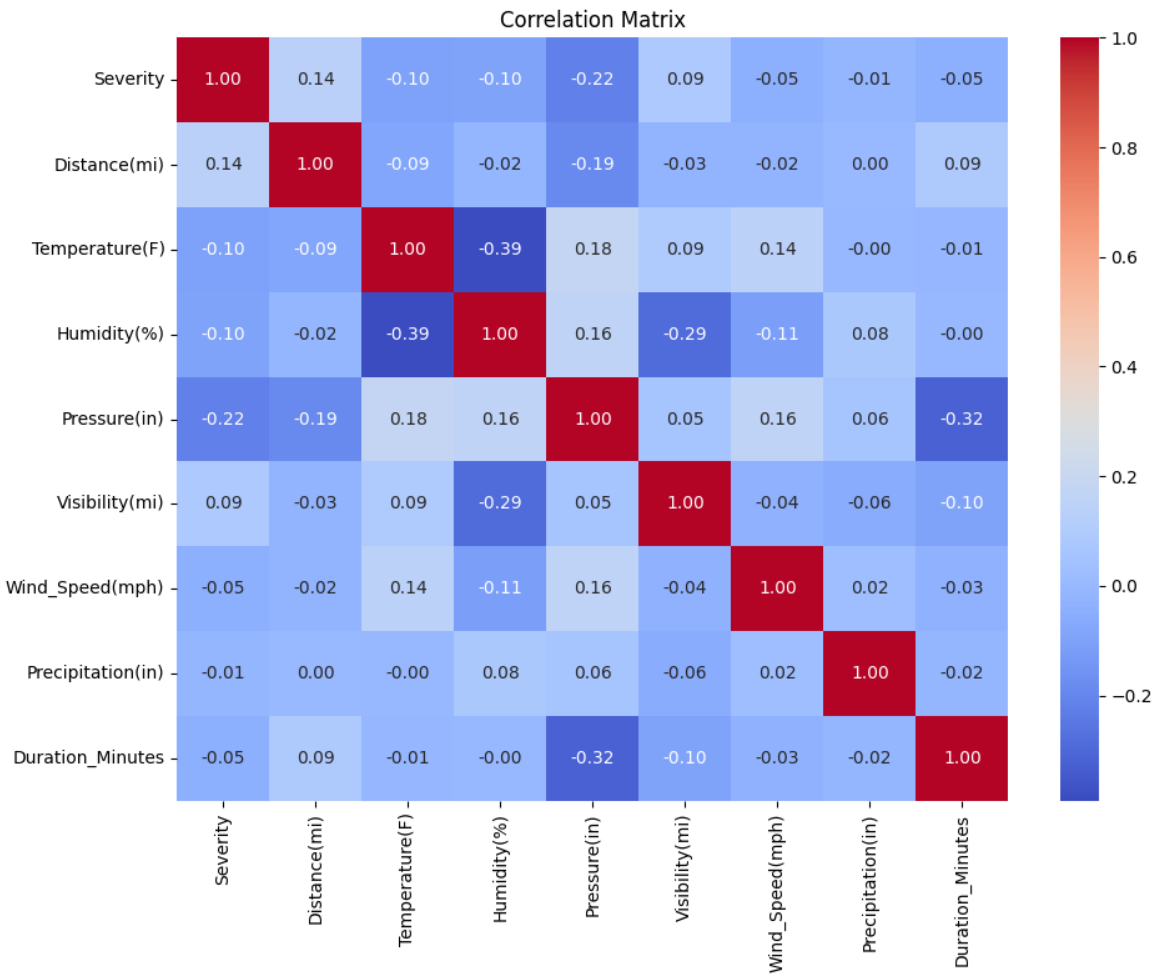
The total counts are similar; however, the types are different OK is heavily dominated in severity 2 accidents, on the other hand CO sees a larger proportion of Severity 3 and slightly more Severity 4 accidents, Crashes in CO tend to be more severity on average.



Severity distribution the majority is classified as severity 2, severity 1 (least impact) and severity 4 (highest impact) is rare.



CO shows a massive rate of accidents at its morning rush hours, 7am-8am, while OK is slightly different where the state's peak hours in the afternoon 4pm-5pm.



The matrix shows that there are no strong linear correlations between accident severity and numerical variables (like weather or distance).